

Additive Models

Suppose we have a response y and predictors $x_1, \dots, x_p$. A linear model takes the form:

$$y = \beta_0 + \sum_{j=1}^p \beta_j X_j + \varepsilon$$

We can include transformations and combinations of the predictors among the x s, so this model can be very flexible. However, it can often be difficult to find a good model, given the wide choice of transformations available. We can try a systematic approach of fitting a family of transformations. For example, we can try polynomials of the predictors, but particularly if we include interactions, the number of terms becomes very large, perhaps greater than the sample size. Alternatively, we can use more interactive and graphical approaches that reward intuition over brute force. However, this requires some skill and effort on the part of the analyst. It is easy to miss important structure; a particular difficulty is that the methods only consider one variable at a time, when the secret to finding good transformations may require that variables be considered simultaneously.

We might try a nonparametric approach by fitting:

$$y = f(x_1, \dots, x_p) + \varepsilon$$

This avoids the necessity of parametric assumptions about the form of the function f , but for p bigger than two or three, it is simply impractical to fit such models due to large sample size requirements, as discussed at the end of the previous chapter.

A good compromise between these extremes is the *additive model*:

$$y = \beta_0 + \sum_{j=1}^p f_j(X_j) + \varepsilon$$

where the f_j are smooth arbitrary functions. Additive models were introduced by Stone (1985).

Additive models are more flexible than the linear model, but still interpretable since the functions f_j can be plotted to give a sense of the marginal relationship between the predictor and the response. Of course, many linear models discovered during a data analysis take an additive form where the transformations are determined in an ad hoc manner by the analyst. The advantage of the additive model approach is that the best transformations are determined simultaneously and without parametric assumptions regarding their form.

In its basic form, the additive model will do poorly when strong interactions exist. In this case we might consider adding terms like $f_{ij}(x_i, x_j)$ or even $f_{ij}(x_i, x_j)$ if there is sufficient data. Categorical variables can be easily accommodated within the model using the usual regression approach. For example:

$$y = \beta_0 + \sum_{j=1}^p f_j(X_j) + Z\gamma + \varepsilon$$

where Z is the design matrix for the variables that will not be modeled additively, where some may be quantitative and others qualitative. The γ are the associated regression parameters. We can also have an interaction between a factor and a continuous predictor by fitting a different function for each level of that factor. For example, we might have f_{male} and f_{female} .

There are several different ways of fitting additive models in R. The `gam` package originates from the work of Hastie and Tibshirani (1990). The `mgcv` package is part of the recommended suite that comes with the default installation of R and is based on methods described in Wood (2000). The `gam` package allows more choice in the smoothers used while the `mgcv` package has an automatic choice in the amount of smoothing as well as wider functionality. The `gss` package of Gu (2002) takes a spline-based approach.

The fitting algorithm depends on the package used. The *backfitting* algorithm is used in the `gam` package. It works as follows:

1. We initialize by setting $\beta_0 = \bar{y}$ and $f_j(x) = \hat{\beta}_j x$ where $\hat{\beta}$ is some initial estimate, such as the least squares, for $j = 1, \dots, p$.
2. We cycle $j = 1, \dots, p, 1, \dots, p, 1, \dots$

$$f_j = S(x_j, y - \beta_0 - \sum_{i \neq j} f_i(X_i))$$

where $S(x, y)$ means the smooth on the data (x, y) . The choice of S is left open to the user. It could be a nonparametric smoother like splines or loess, or it could be a parametric fit, say linear or polynomial. We can even use different smoothers on different predictors with differing amounts of smoothing.

The algorithm is iterated until convergence. Hastie and Tibshirani (1990) show that convergence is assured under some rather loose conditions. The term $y - \beta_0 - \sum_{i \neq j} f_i(x_i)$ is a partial residual — the result of fitting everything except x_j , making the connection to linear model diagnostics.

The `mgcv` package employs a penalized smoothing spline approach. Suppose we represent $f_j(x) = \sum_i \beta_i \phi_i(x)$ for a family of spline basis functions, ϕ_i . We impose a penalty $\int [f_j''(x)]^2 dx$ which can be written in the form $\beta_j^T S_j \beta_j$, for a suitable matrix S_j that depends on the choice of basis. We then maximize:

$$\log L(\beta) - \sum_j \lambda_j \beta_j^T S_j \beta_j$$

where $L(\beta)$ is likelihood with respect to β and the λ_j s control the amount of smoothing for each variable. Generalized cross-validation (GCV) is used to select the λ_j s.

15.1 Modeling Ozone Concentration

We illustrate the methodology in this chapter and some of the following chapters using some data on air pollution from the Los Angeles area in 1976. The response is `O3`, the atmospheric ozone concentration on a particular day. To simplify matters, we will initially reduce the predictors to just three: temperature measured at El Monte, `temp`, inversion base height at LAX, `ibh`, and inversion top temperature at LAX, `ibt`. A number of cases with missing variables have been removed for simplicity. The data were first presented by Breiman and Friedman (1985). We might start with a look at how the response is related to each of the predictors as seen in Figure 15.1. The loess fitted line and confidence band are added using the `geom_smooth()` command. We can see a somewhat nonlinear relationship in all three cases.

```
data(ozone, package="faraway")
ggplot(ozone, aes(x=temp, y=O3)) + geom_point(size=1) + geom_smooth()
ggplot(ozone, aes(x=ibh, y=O3)) + geom_point(size=1) + geom_smooth() +
  ↪ theme(axis.text.x = element_text(angle = 90))
ggplot(ozone, aes(x=ibt, y=O3)) + geom_point(size=1) + geom_smooth()
```

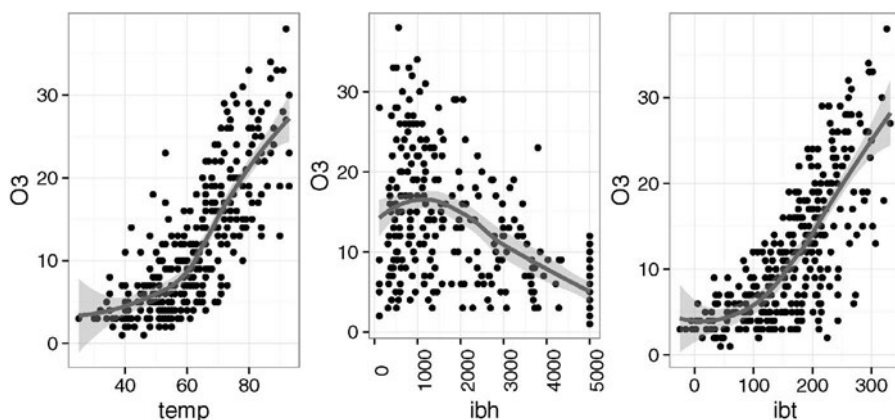


Figure 15.1 Ozone concentration and three predictors. Loess fitted line and confidence band are shown.

These plots show only the raw relationship between the predictors and response without allowance for the effect of other predictors. We can see that, for example, `O3` increases with `ibt` but would this remain true if we adjusted for the effect of `temp` and `ibh`? A first attempt at doing this suggests a linear model, useful at least for reference purposes:

```
olm <- lm(O3 ~ temp + ibh + ibt, ozone)
summary(olm)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-7.727982	1.621662	-4.77	2.8e-06
temp	0.380441	0.040158	9.47	< 2e-16
ibh	-0.001186	0.000257	-4.62	5.5e-06
ibt	-0.005821	0.010179	-0.57	0.57

`n = 330, p = 4, Residual SE = 4.748, R-Squared = 0.65`

Note that `ibt` is not significant in this model. Although we can determine the individual relationships between the predictors and the response by examining the regression coefficients, it can be helpful to view these effects graphically. One can use the `termplot` function but the `effects` package due to Fox (2003) has more comprehensive capabilities for achieving this functionality. We fix the other predictors at some typical value (the mean by default) and allow only the chosen predictor to vary. We plot the resulting prediction. Plots of these predictions are shown in Figure 15.2 for each of the predictors. We have added the partial residuals to the display. The dashed line represents a smooth fit to these residuals and gives a suggestion of the extent of nonlinearity.

```
library(effects)
plot(Effect("temp", olm, partial.residuals=TRUE))
plot(Effect("ibh", olm, partial.residuals=TRUE))
plot(Effect("ibt", olm, partial.residuals=TRUE))
```

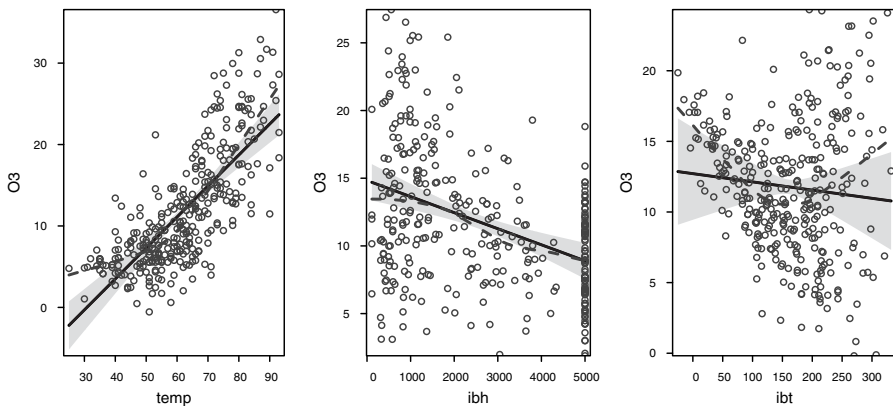


Figure 15.2 *Effects of predictors in a linear model for Ozone concentration.*

These plots show us how much the predicted response will vary over the observed range of the predictor and so are helpful in assessing the magnitude and direction of the effect. We can see from these plots that `ibt` does not have much effect after adjusting for the other two predictors, although there is some suggestion of a quadratic effect.

15.2 Additive Models Using `mgcv`

A method of fitting additive models is provided by the `mgcv` package of Wood (2006). Splines are the default choice of smoother with the appropriate amount of smoothing chosen automatically, unless we intervene.

```
library(mgcv)
ammgcv <- gam(O3 ~ s(temp)+s(ibh)+s(ibt), data=ozone)
summary(ammgcv)
```

```
Parametric coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   11.776      0.238    49.4   <2e-16
```

Approximate significance of smooth terms:

	edf	Ref.df	F	p-value
s(temp)	3.39	4.26	20.55	7.7e-16
s(ibh)	4.17	5.08	7.34	1.4e-06
s(ibt)	2.11	2.73	1.61	0.19

R-sq.(adj) = 0.708 Deviance explained = 71.7%

GCV = 19.346 Scale est. = 18.72 n = 330

The intercept is the only parametric coefficient in this model because all the predictor terms have smooths. We can compute the equivalent degrees of freedom by an analogy to linear models. For linear smoothers, the relationship between the observed and fitted values may be written as $\hat{y} = Py$. The trace of P then estimates the effective number of parameters. For example, in linear regression, the projection matrix is $X(X^T X)^{-1} X^T$ whose trace is equal to the rank of X or the number of identifiable parameters. This notion can be used to obtain the degrees of freedom for additive models. The column marked `Ref.df` is a modified computation of the degrees of freedom which is more appropriate for use in test statistics.

Since we have sums of squares and degrees of freedom, we can compute F -statistics in the same way as linear models. However, the F -statistics quoted in the summary output have been modified to produce somewhat better statistical properties. The p -values are computed from these F -statistics and degrees of freedom although we cannot claim the null distributions are exactly F -distributed. Usually, they are good approximations. We see that the R^2 , which in this case is called the “Deviance explained,” is somewhat higher than in the `lm` fit.

We can also examine the transformations used:

```
plot(ammgcv, residuals=TRUE, select=1)
plot(ammgcv, residuals=TRUE, select=2)
plot(ammgcv, residuals=TRUE, select=3)
```

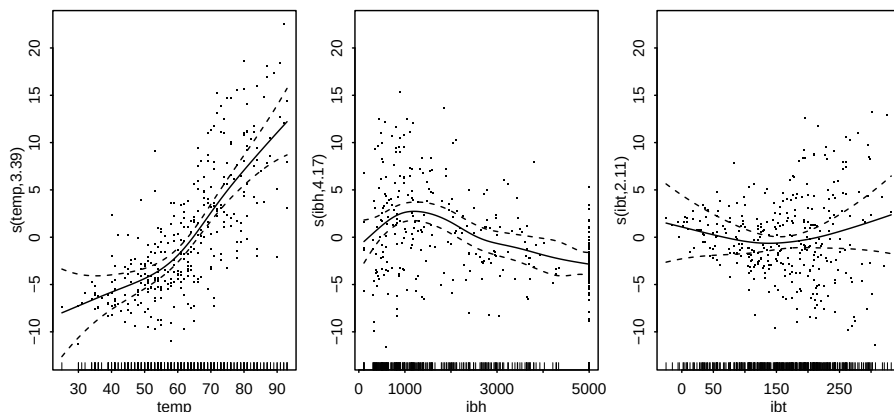


Figure 15.3 Transformation functions for the model fit by `mgcv`. Note how the same scale has been deliberately used on all three plots. This allows us to easily compare the relative contribution of each variable.

We see that the transformations, as seen in Figure 15.3, are nonlinear. The chosen transformation for temperature is roughly piecewise linear with a change in the slope around 60. We see a similar piecewise linear relationship with a change at about 1000 for *ibh*. A deeper understanding of the application might lead to some interpretation of these changepoints. We have asked for the residuals to be shown on the plot, in keeping with our previous plots, as this adds to our ability to judge the choice of smoothing. We can see the degree of smoothing appears appropriate as there is no obvious finer structure. Note that the label on the vertical axis gives the degrees of freedom for the smooth which will correspond to the shape of the fit. Something close to linear will have degrees of freedom close to one.

The confidence bands are helpful in judging the significance of features seen in the plots. We might judge whether a straight line could be drawn entirely between the two bands. This might suggest that a linear term would be sufficient. In the case of *ibt*, we see that a horizontal line at zero would fit between the bands. This suggests this predictor has no marginal effect on the response.

We might also be interested in whether there really is a change in the trend for temperature. We test this supposition with a linear term in temperature and then make the *F*-test:

```
am1 <- gam(O3 ~ s(temp)+s(ibh),data=ozone)
am2 <- gam(O3 ~ temp+s(ibh),data=ozone)
anova(am2,am1,test="F")
```

Analysis of Deviance Table

```
Model 1: O3 ~ temp + s(ibh)
Model 2: O3 ~ s(temp) + s(ibh)
```

	Resid. Df	Resid. Dev	Df	Deviance	F	Pr(>F)
1	324	6950				
2	321	6054	2.7	896	17.6	7.9e-10

The *p*-value is only approximate, but we can be confident there is a change in the trend for temperature.

You can include functions of two variables with *mgcv*. This remains within the framework of the additive model. For example, suppose we suspect that there is an interaction between temperature and IBH. We must decide on the appropriate bivariate smoothing method. In some cases, the two variables are measured in the same units. For example, one might measure location on the north-south axis and another the location on the east-west axis. In this *isotropic* case, it is appropriate to use the same degree of smoothing in both variables. But in our example, temperature and inverse base height (IBH) are measured on different scales and are thus *anisotropic*. In this case, we will likely need a different amount of smoothing the two directions. This can be achieved using *tensor product* smooth:

```
amint <- gam(O3 ~ te(temp,ibh)+s(ibt),data=ozone)
summary(amint)
```

Parametric coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	11.776	0.239	49.4	<2e-16

Approximate significance of smooth terms:

	edf	Ref.df	F	p-value
te(temp,ibh)	10.49	13.08	12.96	<2e-16

```
s(ibt)          1.04   1.08   0.47   0.51
```

```
R-sq.(adj) = 0.707   Deviance explained = 71.8%
GCV = 19.526   Scale est. = 18.784   n = 330
```

We see that the combined temperature and IBH term are clearly significant with the `ibt` term remaining redundant. This output does not tell us whether the interaction between the two terms adds anything to the model although the very small increase in the deviance explained suggests that it does not.

We can test the significance of the interaction:

```
anova(ammgcv, amint, test="F")
```

```
Analysis of Deviance Table
```

```
Model 1: O3 ~ s(temp) + s(ibh) + s(ibt)
```

```
Model 2: O3 ~ te(temp, ibh) + s(ibt)
```

	Resid. Df	Resid. Dev	Df	Deviance	F	Pr(>F)
1	319	5978				
2	317	5963	1.86	14.6	0.42	0.64

The large p -value confirms the insignificance of the interaction term. We can be satisfied with just the two univariate additive terms. In more interesting cases, we would be eager to visualize the interaction as seen in two different ways in Figure 15.4:

```
plot(amint, select=1)
```

```
vis.gam(amint, theta=-45, color="gray")
```

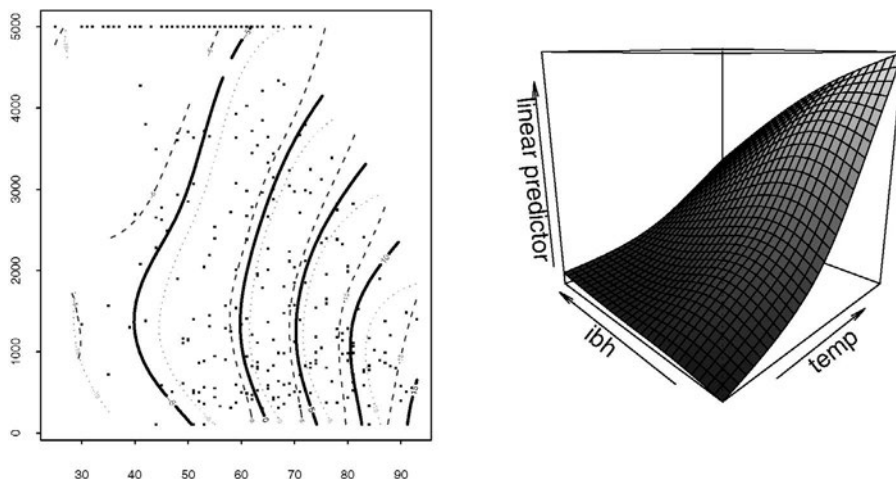


Figure 15.4 The bivariate contour plot for temperature and `ibh` is shown in the left panel. The right panel shows a perspective view of the information on the left panel.

The contours appear almost parallel. The perspective view also suggests no interaction.

One use for additive models is as an exploratory tool for standard parametric regression modeling. We can use the fitted functions to help us find suitable simple transformations of the predictors. One idea here is to model the `temp` and `ibh` ef-

fects using piecewise linear regression (also known as “broken stick” or segmented regression). We define the right and left “hockey-stick” functions:

```
rhs <- function(x,c) ifelse(x > c, x-c, 0)
lhs <- function(x,c) ifelse(x < c, c-x, 0)
```

and now fit a parametric model using cutpoints of 60 and 1000 for temp and ibh, respectively. We pick the cutpoints using the plots:

```
olm2 <- lm(O3 ~ rhs(temp,60)+lhs(temp,60)+rhs(ibh,1000)+lhs(ibh,1000),
  ↪ ozone)
summary(olm2)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	11.603832	0.622651	18.64	< 2e-16
rhs(temp, 60)	0.536441	0.033185	16.17	< 2e-16
lhs(temp, 60)	-0.116173	0.037866	-3.07	0.0023
rhs(ibh, 1000)	-0.001486	0.000198	-7.49	6.7e-13
lhs(ibh, 1000)	-0.003554	0.001314	-2.71	0.0072

n = 330, p = 5, Residual SE = 4.342, R-Squared = 0.71

Compare this model to the first linear model we fit to this data. The fit is better and about as good as the additive model fit. It is unlikely we could have discovered these transformations without the help of the intermediate additive models. Furthermore, the linear model has the advantage that we can write the prediction formula in a compact form.

We can use additive models for building a linear model as above, but they can be used for inference in their own right. For example, we can predict new values with standard error:

```
predict(ammgcv, data.frame(temp=60, ibh=2000, ibt=100), se=T)
$fit
[1] 11.013

$se.fit
[1] 0.97278
```

If we try to make predictions for predictor values outside the original range of the data, we will need to linearly extrapolate the spline fits. This is dangerous for all the usual reasons:

```
predict(ammgcv, data.frame(temp=120, ibh=2000, ibt=100), se=T)
$fit
[1] 35.511

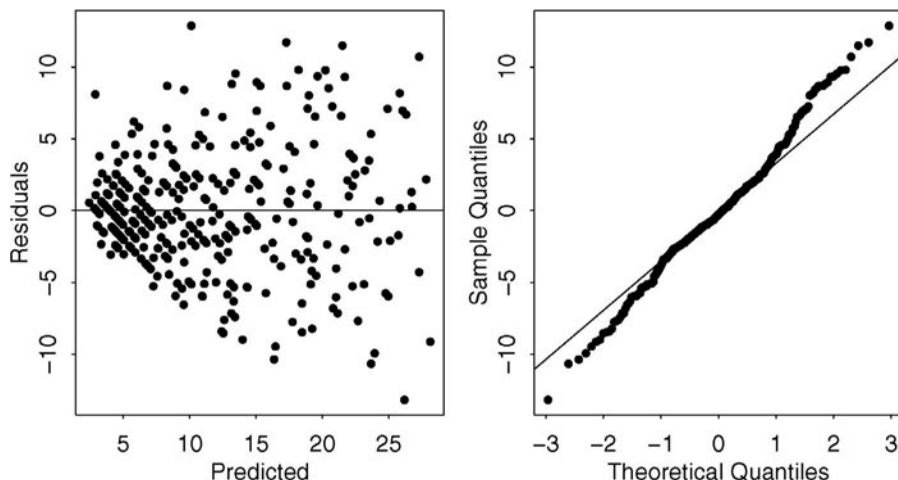
$se.fit
[1] 5.7261
```

We see that the standard error is much larger although this likely does not fully reflect the uncertainty.

We should also check the usual diagnostics:

```
plot(residuals(ammgcv)~predict(ammgcv), xlab="Predicted", ylab="
  ↪ Residuals")
abline(h=0)
qqnorm(residuals(ammgcv), main="")
qqline(residuals(ammgcv))
```

We can see in Figure 15.5 there is some nonconstant variance. There are also somewhat long tails for the residuals.

Figure 15.5 *Residual plots for the additive model.*

Now let's see the model for the full dataset. We found that the `ibh` and `ibt` terms were insignificant and so we removed them:

```
amred <- gam(O3 ~ s(vh)+s(wind)+s(humidity)+s(temp)+s(dpg)+ s(vis)+s(
  ↪ doy),data=ozone)
summary(amred)
```

Parametric coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	11.776	0.201	58.7	<2e-16

Approximate significance of smooth terms:

	edf	Ref.df	F	p-value
s(vh)	1.00	1.00	20.50	8.4e-06
s(wind)	1.00	1.00	6.56	0.01091
s(humidity)	1.00	1.00	14.61	0.00016
s(temp)	5.77	6.92	12.66	2.1e-14
s(dpg)	3.31	4.20	14.11	6.4e-11
s(vis)	2.22	2.77	7.20	0.00021
s(doy)	4.07	5.13	20.80	< 2e-16

R-sq.(adj) = 0.793 Deviance explained = 80.5%

GCV = 14.113 Scale est. = 13.285 n = 330

We will compare this to the results of different modeling approaches that we will present later. We can see that we achieve a good fit with an R^2 of 80.5%, but at the cost of using effectively 19.4 (sum the df) parameters including the intercept.

Also for future reference, here is the linear model with all insignificant terms removed:

```
alm <- lm(O3 ~ vis+doy+ibt+humidity+temp,data=ozone)
summary(alm)
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-10.01786	1.65306	-6.06	3.8e-09
vis	-0.00820	0.00369	-2.22	0.027

doy	-0.01020	0.00245	-4.17	3.9e-05
ibt	0.03491	0.00671	5.21	3.4e-07
humidity	0.08510	0.01435	5.93	7.7e-09
temp	0.23281	0.03607	6.45	4.0e-10

n = 330, p = 6, Residual SE = 4.430, R-Squared = 0.7

We can see that the fit is substantially worse, but uses only six parameters. Of course, we may be able to improve this fit with some manual data analysis. We could look for good transformations and check for outliers and influential points. However, since we want to compare different modeling techniques, we want to avoid making subjective interventions for the sake of a fair comparison.

15.3 Generalized Additive Models

In generalized linear models:

$$\eta = X\beta \quad EY = \mu \quad g(\mu) = \eta \quad \text{Var}(Y) \propto V(\mu)$$

The approach is readily extended to additive models to form generalized additive models (GAM). We replace the linear predictor with

$$\eta = \beta_0 + \sum_{j=1}^p f_j(X_j)$$

In the `mgcv` package, the f_j are represented by splines. These splines have coefficients that are just more parameters that can be estimated using the likelihood approach.

The ozone data has a response with relatively small integer values. Furthermore, the diagnostic plot in Figure 15.5 shows nonconstant variance. This suggests that a Poisson response might be suitable. We fit this using:

```
gammgcv <- gam(O3 ~ s(temp)+s(ibh)+s(ibt), family=poisson, scale=-1,
  ↪ data=ozone)
summary(gammgcv)
```

Parametric coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.293	0.023	99.5	<2e-16

Approximate significance of smooth terms:

	edf	Ref.df	F	p-value
s(temp)	3.82	4.74	16.82	4.2e-14
s(ibh)	3.74	4.57	10.54	1.2e-08
s(ibt)	1.35	1.62	0.57	0.53

R-sq.(adj) = 0.712 Deviance explained = 72.9%
 GCV = 1.5062 Scale est. = 1.4573 n = 330

We have set `scale=-1` because negative values for this parameter indicate that the dispersion should be estimated rather than fixed at one. Since we do not truly believe the response is Poisson, it seems wise to allow for overdispersion. The default of not specifying `scale` would fix the dispersion at one. We see that the estimated dispersion is indeed somewhat bigger than one. We see that IBT is not significant. We can check the transformations on the predictors as seen in Figure 15.6:

```
plot(gammgcv, residuals=TRUE, select=1)
plot(gammgcv, residuals=TRUE, select=2)
plot(gammgcv, residuals=TRUE, select=3)
```

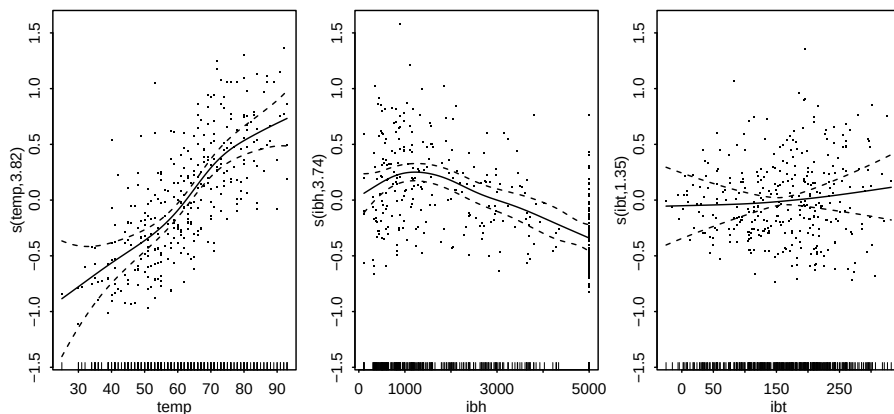


Figure 15.6 Transformation on the predictors for the Poisson GAM.

We see that the selected transformations are quite similar to those observed previously.

There are natural extensions to additive modeling for all the response types we have considered earlier in this book.

15.4 Alternating Conditional Expectations

In the additive model:

$$y = \alpha + \sum_{j=1}^p f_j(X_j) + \varepsilon$$

but in the transform-both-sides (TBS) model:

$$\theta(y) = \alpha + \sum_{j=1}^p f_j(X_j) + \varepsilon$$

For example, $y = e^{x_1 + \sqrt{x_2}}$ cannot be modeled well by additive models, but can if we transform both sides: $\log y = x_1 + \sqrt{x_2}$. This fits within the TBS model framework. A more complicated alternative approach would be nonlinear regression. One particular way of fitting TBS models is *alternating conditional expectation* (ACE) which is designed to minimize $\sum_i (\theta(y_i) - \sum f_j(x_{ij}))^2$. Distractingly, this can be trivially minimized by setting $\theta = f_j = 0$ for all j . To avoid this solution, we impose the restriction that the variance of $\theta(y)$ be one. The fitting proceeds using the following algorithm:

1. Initialize:

$$\theta(y) = \frac{y - \bar{y}}{SD(y)} \quad f_j = \hat{\beta}_j x_j \quad j = 1, \dots, p$$

2. Cycle:

$$f_j = S(x_j, \theta(y) - \sum_{i \neq j} f_i(x_i))$$

$$\theta = S(y, \sum_j f_j(x_j))$$

where $S(x, y)$ represents a smoother returning a function on (x, y) data. Renormalize at the end of each cycle:

$$\theta(y) \leftarrow \frac{\theta(y) - \overline{\theta(y)}}{SD(\theta(y))}$$

We repeat until convergence. ACE is comparable to the additive model, except now we allow transformation of the response as well. In principle, you can use any reasonable smoother S , but the original smoother used was the supersmoother. This cannot be easily changed in the R software implementation.

For our example, we start with the same three predictors in the `ozone` data:

```
x <- ozone[,c("temp", "ibh", "ibt")]
library(acepack)
acefit <- ace(x, ozone$O3)
```

Note that the `ace` function interface is quite rudimentary as we must give it the X matrix explicitly. The function returns the component `ty` which contains $\theta(y)$ and `tx` which is a matrix whose columns contain the $f_j(x_j)$. We can get a sense of how well these transformations work by fitting a linear model that uses the transformed variables. We know that the intercept is zero so we exclude it to get cleaner output:

```
summary(lm(ty ~ tx-1, acefit))
```

	Estimate	Std. Error	t value	Pr(> t)
txtemp	0.9676	0.0508	19.04	<2e-16
txibh	1.1801	0.1358	8.69	<2e-16
txibt	1.3712	0.5116	2.68	0.0077

`n = 330, p = 3, Residual SE = 0.526, R-Squared = 0.73`

All three transformed predictors are strongly significant and the fit is superior to the original model. The R^2 for the comparable additive model was 0.703. So the additional transformation of the response did improve the fit. Now we examine the transforms on the response and the three predictors:

```
plot(ozone$O3, acefit$ty, xlab="O3", ylab=expression(theta(O3)))
plot(x[,1], acefit$tx[,1], xlab="temp", ylab="f(temp)")
plot(x[,2], acefit$tx[,2], xlab="ibh", ylab="f(ibh)")
plot(x[,3], acefit$tx[,3], xlab="ibt", ylab="f(ibt)")
```

See Figure 15.7. The transform on the response is close to, but not quite, linear. The transformations on `temp` and `ibh` are similar to those found by the additive model. The transformation for `ibt` looks implausibly rough in some parts.

Now let's see how we do on the full data:

```
x <- ozone[, -1]
acefit <- ace(x, ozone$O3)
summary(lm(acefit$ty ~ acefit$tx-1))
```

	Estimate	Std. Error	t value	Pr(> t)
acefit\$txvh	1.172	0.385	3.05	0.0025

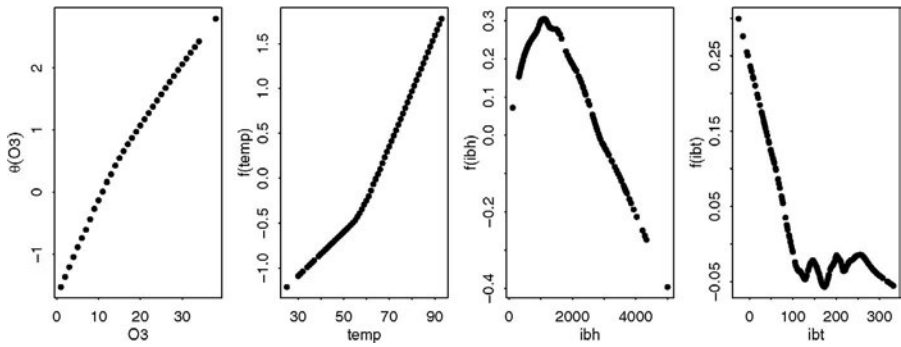


Figure 15.7 ACE transformations: the first panel shows the transformation on the response while the remaining three show the transformations on the predictors.

acefit\$txwind	1.074	0.404	2.66	0.0083
acefit\$txhumidity	0.651	0.245	2.66	0.0083
acefit\$txtemp	0.916	0.123	7.43	1.0e-12
acefit\$txibh	1.351	0.436	3.10	0.0021
acefit\$txdpg	1.322	0.167	7.92	4.0e-14
acefit\$txibt	0.926	0.196	4.71	3.7e-06
acefit\$txvis	1.386	0.230	6.03	4.5e-09
acefit\$txdoy	1.284	0.110	11.72	< 2e-16

n = 330, p = 9, Residual SE = 0.408, R-Squared = 0.84

A very good fit, but we must be cautious. Notice that all the predictors are strongly significant. This might be a reflection of reality or it could just be that the ACE model is overfitting the data by using implausible transformations as seen on the `ibt` variable above. Another problem is in constructing new predictions from fresh inputs. The nature of the smoothing used does not make it obvious how this can be achieved.

ACE can be useful in searching for good transformations while building a linear model. We might examine the fitted transformations as seen in Figure 15.7 to suggest appropriate parametric forms. More caution is necessary if the model is to be used in its own right, because of the tendency to overfit.

An alternative view of ACE is to consider the problem of choosing θ and f_j 's such that $\theta(Y)$ and $\sum_j f_j(X_j)$ are maximally correlated. ACE solves this problem. For this reason, ACE can be viewed as a correlation method rather than a regression method.

The canonical correlation method is an ancestor to ACE. Given two sets of random variables $X_1, \dots, X_m$ and $Y_1, \dots, Y_n$, we find unit vectors a and b such that:

$$\text{corr}(a^T X, b^T Y)$$

is maximized. One generalization of canonical correlation is to allow some of the X 's and Y 's to be power transforms of the original variables; this results in a parametric form of ACE. For example:

```
y <- cbind(ozone$O3, ozone$O3^2, sqrt(ozone$O3))
x <- ozone[, c("temp", "ibh", "ibt")]
cancor(x, y)
$cor
[1] 0.832346 0.217517 0.016908
```

```
$xcoef
      [,1]      [,2]      [,3]
temp -3.4951e-03  3.6335e-03 -6.7913e-03
ibh   1.3667e-05 -5.2054e-05 -5.2243e-06
ibt   1.6744e-04 -1.7384e-03  1.2436e-03
```

```
$ycoef
      [,1]      [,2]      [,3]
[1,] -0.00390830 -0.00539076 -0.1802230
[2,]  0.00009253 -0.00044172  0.0022167
[3,] -0.03928664  0.12068982  0.7948130
```

We see that it is possible to obtain a correlation of 0.832 by taking particular linear combinations of $O3$, $O3^2$ and $\sqrt{O3}$ with the three predictors. The other two orthogonal combinations are not of interest to us here. Remember that R^2 is the correlation squared in a simple linear model and $0.832^2 = 0.692$, so this is not a particularly competitive fit.

There are some oddities about ACE. For a single predictor, ACE is symmetric in X and Y , which is not the usual situation in regression. Furthermore, ACE does not necessarily reproduce the true model. Consider the population form of the problem and take $Y = X + \epsilon$ and $\epsilon \sim N(0, 1)$ and $X \sim U(0, 1)$, then $E(Y|X) = X$ but $E(X|Y) \neq Y$ which is not what one might expect, because f and θ will not both be identity transformations as the model might suggest.

15.5 Additivity and Variance Stabilization

Additivity and variance stabilization (AVAS) is another TBS model and is quite similar to ACE. We choose the f_j to optimize the fit, but we also want constant variance for the response:

$$\text{var}[\theta(Y) | \sum_{j=1}^p f_j(X_j)] = \text{constant}$$

So we choose the f_j 's to get a good additive fit and choose the θ to get constant variance.

Here is how the method of fitting θ works: suppose $\text{Var}(Y) \equiv V(Y)$ is not constant. We transform to constancy by:

$$\theta(t) = \int_0^t \frac{d\mu}{\sqrt{V(\mu)}}$$

We use data to estimate $V(y)$, then get θ . The purpose of the AVAS method is to obtain additivity and variance stabilization and not necessarily to produce the best possible fit. We demonstrate its application on the `ozone` data:

```
avasfit <- avas(x, ozone$O3)
```

Plot the transformations selected:

```
plot(ozone$O3, avasfit$ty, xlab="O3", ylab=expression(theta(O3)))
plot(x[,1], avasfit$tx[,1], xlab="temp", ylab="f(temp)")
plot(x[,2], avasfit$tx[,2], xlab="ibh", ylab="f(ibh)")
plot(x[,3], avasfit$tx[,3], xlab="ibt", ylab="f(ibt)")
```

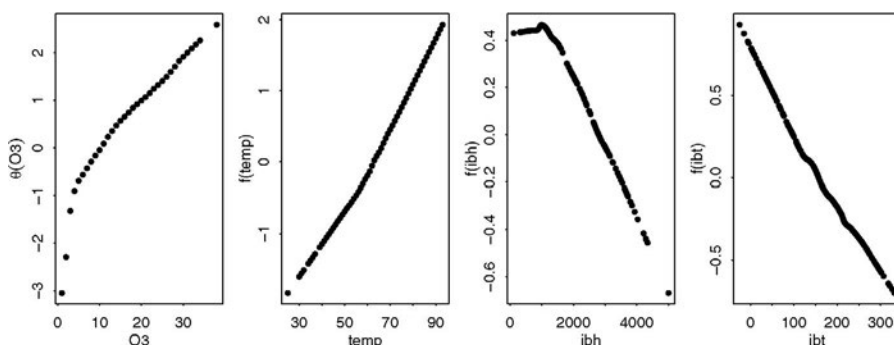


Figure 15.8 AVAS transformations — the first panel shows the transformation on the response while the remaining three show the transformations on the predictors.

See Figure 15.8. It would be convenient if the transformation on the response matched a simple functional form. We see if this is possible. We need to sort the response to get the line plots to work:

```
i <- order(ozone$O3)
plot(ozone$O3[i], avasfit$ty[i], type="l", xlab="O3", ylab=expression(
  ↪ theta(O3)))
gs <- lm(avasfit$ty[i] ~ sqrt(ozone$O3[i]))
lines(ozone$O3[i], gs$fit, lty=2)
gl <- lm(avasfit$ty[i] ~ log(ozone$O3[i]))
lines(ozone$O3[i], gl$fit, lty=5)
```

See the left panel of Figure 15.9. We have shown the square-root fit as a dotted line and log fit as a dashed line. Neither one fits well across the whole range. Now look at the overall fit:

```
lmod <- lm(avasfit$ty ~ avasfit$tx - 1)
summary(lmod)
```

	Estimate	Std. Error	t value	Pr(> t)
avasfit\$txtemp	0.9006	0.0746	12.08	< 2e-16
avasfit\$txibh	0.7950	0.1091	7.29	2.4e-12
avasfit\$txibt	0.5637	0.2389	2.36	0.019

n = 330, p = 3, Residual SE = 0.562, R-Squared = 0.69

The fit is not so good, but check the diagnostics:

```
plot(predict(lmod), residuals(lmod), xlab="Fitted", ylab="Residuals")
```

The plot is shown in the right panel of Figure 15.9.

AVAS does not optimize the fit; it trades some of the optimality in order to obtain constant variance. Whether this is a good trade depends on how much relative value you put on the accuracy of point predictions and accurate estimation of the standard error of prediction. In other words, is it more important to try to be right or to know how much you are wrong? The choice will depend on the application. An alternative

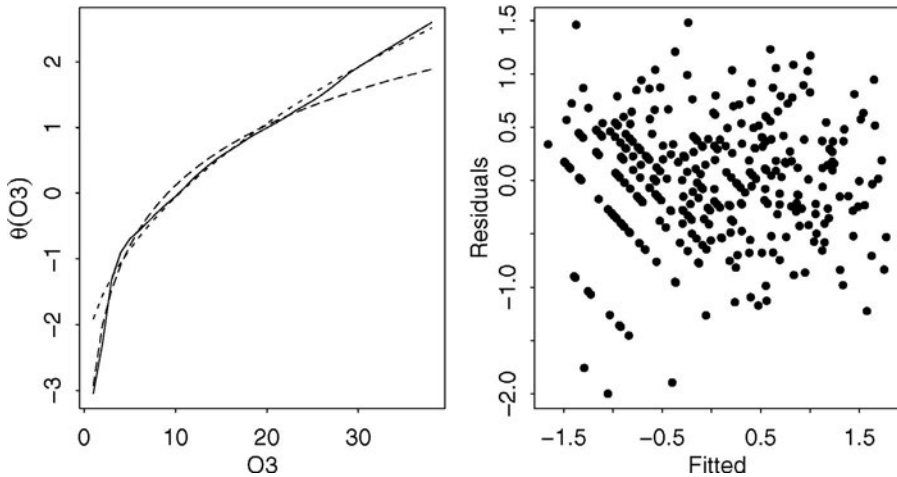


Figure 15.9 The left panel checks for simple fits to the AVAS transformation on the response given by the solid line. The log fit is given by the dashed line while the square-root fit is given by the dotted line. The right panel shows the residuals vs. fitted values plot for the AVAS model.

approach is to accept the nonconstant variance but try to model it and modify the uncertainty expressed in predictions accordingly. This sophistication in modeling is not always possible to achieve within some of the nonparametric regression frameworks.

15.6 Generalized Additive Mixed Models

The generalized additive mixed model (GAMM) manages to combine the three major themes of this book. The response can be nonnormal from the exponential family of distributions. The error structure can allow for grouping and hierarchical arrangements in the data. Finally we can allow for smooth transformations of the response. We demonstrate this method on the epilepsy data from Section 13.5:

```
data(epilepsy, package="faraway")
egamm <- gamm(seizures ~ offset(timeadj) + treat*expind+s(age), family
  ↪ =poisson, random=list(id=~1), data=epilepsy, subset=(id!=49))
summary(egamm$gam)
```

Parametric coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-4.8392	0.1434	-33.74	<2e-16
treat	-0.0104	0.1999	-0.05	0.9584
expind	4.7255	0.0758	62.38	<2e-16
treat:expind	-0.3024	0.1127	-2.68	0.0077

Approximate significance of smooth terms:

	edf	Ref.df	F	p-value
s(age)	1	1	0.31	0.58


```
R-sq.(adj) = 0.328
Scale est. = 2.5765    n = 290
```

We see that the age effect is not significant. Again the interaction effect is significant which shows, in this case, a beneficial effect for the drug.

15.7 Multivariate Adaptive Regression Splines

Multivariate adaptive regression splines (MARS) were introduced by Friedman (1991). We wish to find a model of the form:

$$\hat{f}(x) = \sum_{j=1}^k c_j B_j(x)$$

where the basis functions, $B_j(x)$, are formed from products of terms of the form $[\pm(x_i - t)]_+^q$. The $[\]_+$ denotes taking the positive part. For $q = 1$, this is sometimes called a hockey-stick function and can be seen in the right panel of Figure 14.5. The $q = 1$ case is the most common choice and one might think this results in a piecewise linear fit, as in Section 14.2. However, the fit is more flexible than this. Consider the product of two hockey-stick functions in one dimension; this forms a quadratic shape. Furthermore, if we form the product of terms in two or more variables, we have an interaction term.

The model building proceeds iteratively. We start with no basis functions. We then search over all variables and possible knotpoints t to find the one basis function that produces the best fit to the data. We now repeat this process to find the next best basis function addition given that the first basis function is already included in the model. We might impose rules on what new basis functions are included. For example, we might disallow interactions or only allow up to two-way interactions. This will enhance interpretability, possibly at the cost of fit. The number of basis functions added determines the overall smoothness of the fit. We can use cross-validation to determine how many basis functions are enough.

When interactions are disallowed, the MARS approach will be a type of additive model. The MARS model building will be iterative, in contrast to the fit using the `gam` function of the `mgcv` package that fits and determines overall smoothness in one step. If a strictly additive model is all that is needed, the `mgcv` approach will typically be more effective. The MARS approach will have a relative advantage when interactions are considered, particularly when there are a large number of variables. Here there will be a large number of potential interactions that cannot be simultaneously entertained. The iterative approach of MARS will be more appropriate here.

We apply the MARS method to the `ozone` dataset. We use the `earth` package which is built on the functionality in the original `mda` package.

```
library(earth)
mmmod <- earth(O3 ~ ., ozone)
summary(mmmod)
```

```
coefficients
(Intercept)      11.01166
h(5890-vh)       -0.01349
h(9-wind)         0.27685
```

```

h(humidity-41)      0.27438
h(humidity-54)     -0.26598
h(temp-58)         0.38519
h(1049-ibh)        -0.00269
h(ibh-1049)        -0.00056
h(dpg-10)          -0.10564
h(150-vis)         0.02336
h(96-doy)          -0.11977
h(doy-96)          0.03885
h(doy-158)         -0.08187

```

Selected 13 of 19 terms, and 8 of 9 predictors

Termination condition: Reached nk 21

Importance: temp, doy, humidity, dpg, ibh, vh, vis, wind, ibt-unused

Number of terms at each degree of interaction: 1 12 (additive model)

GCV 14.573 RSS 4108.1 GRSq 0.77362 RSq 0.80545

The default choice allows only additive (first-order) predictors and chooses the model size using a GCV criterion. The basis functions are expressed in terms of the break-points.

The fit is good in terms of R^2 , but the model size is also larger. It is also an additive model, so we can reasonably compare it to the additive model presented at the end of Section 15.2. That model had an adjusted R^2 of 79.3% using 19.4 parameters.

Let's reduce the model size to that used for previous models. The parameter `nk` controls the maximum number of model terms:

```

mmmod <- earth(O3 ~ ., ozone, nk=7)
summary(mmmod)

```

```

coefficients
(Intercept)      13.07286
h(58-temp)       -0.10623
h(temp-58)        0.49066
h(1049-ibh)      -0.00253
h(ibh-1049)      -0.00142
h(96-doy)        -0.09609
h(doy-96)        -0.01279

```

Selected 7 of 7 terms, and 3 of 9 predictors

Termination condition: Reached nk 7

Importance: temp, ibh, doy, vh-unused, wind-unused, humidity-unused,
dpg-unused, ibt-unused, vis-unused

Number of terms at each degree of interaction: 1 6 (additive model)

GCV 18.284 RSS 5567.8 GRSq 0.71597 RSq 0.73631

This fit is worse, but remember we are disallowing any interaction terms. Now let's allow second-order (two-way) interaction terms. `nk` was chosen to get the same model size as before:

```

mmmod <- earth(O3 ~ ., ozone, nk=7, degree=2)
summary(mmmod)

```

```

coefficients
(Intercept)      10.03251
h(58-temp)       -0.10668
h(temp-58)        0.45593
h(ibh-1049)      -0.00115
h(55-humidity) * h(temp-58) -0.01311
h(humidity-55) * h(temp-58)  0.00599

```

Selected 6 of 7 terms, and 3 of 9 predictors

Termination condition: Reached nk 7

Importance: temp, ibh, humidity, vh-unused, wind-unused, dpg-unused,
ibt-unused, vis-unused, doy-unused

Number of terms at each degree of interaction: 1 3 2

GCV 18.383 RSS 5580.2 GRSq 0.71444 RSq 0.73573

This is a good fit. Compare this with an additive model approach. Since there are nine predictors, this would mean 36 possible two-way interaction terms. Such a model would be complex to estimate and interpret. In contrast, the MARS approach reduces the complexity by carefully selecting the interaction terms.

Now let's see how the terms enter into the model, as seen in Figure 15.10:

```
plotmo(mmod)
```

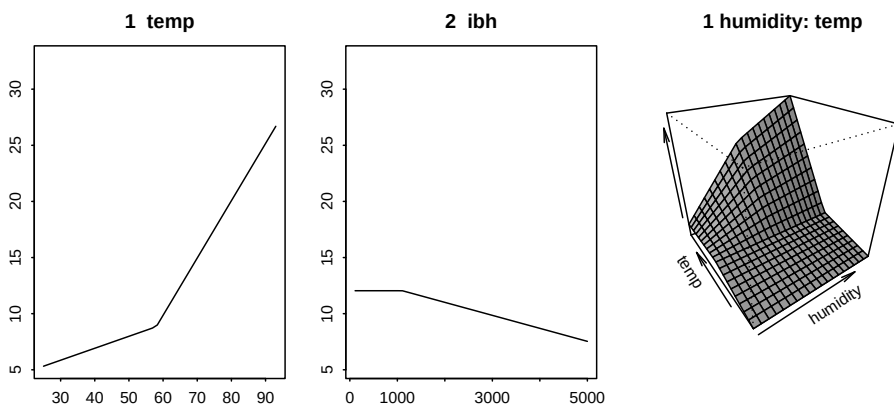


Figure 15.10 Contribution of predictors in the MARS model.

We see similar transformations to those used previously. Now check the diagnostics:

```
plot(mmod, 3)
```

```
plot(mmod, 4)
```

These plots, seen in Figure 15.11, show no problem with normality, but some indication of nonconstant variance.

It is interesting to compare the MARS approach to the univariate version as demonstrated in Figure 14.6. There we used a moderate number of knots in just one dimension while MARS gets by with just a few knots in higher dimensions. The key is to choose the right knots. MARS can be favorably compared to linear regression: it has additional flexibility to find nonlinearity in the predictors in higher dimensions. MARS can also be favorably compared to the tree method discussed in the next chapter: it allows for continuous fits but still maintains good interpretability.

Further Reading: Hastie and Tibshirani (1990) provide the original overview of additive modeling, while Wood (2006) gives a more recent introduction. Gu (2002) presents another approach to the problem. Green and Silverman (1993) show the link

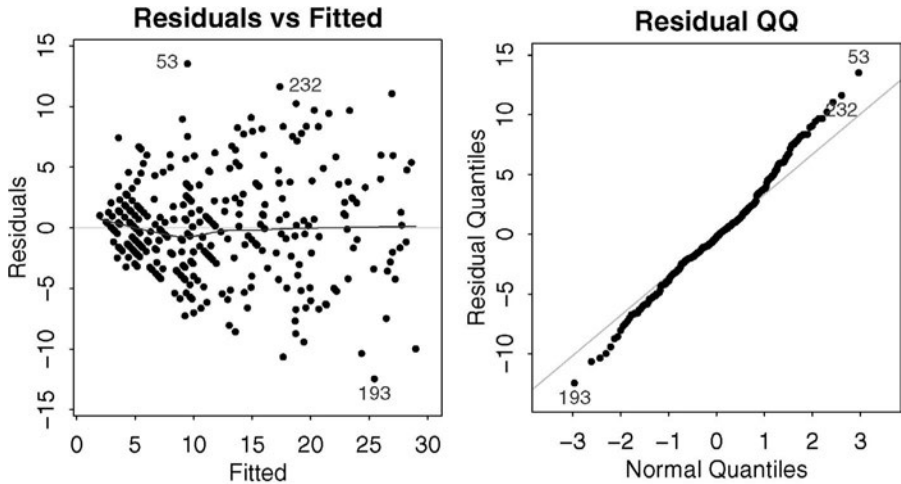


Figure 15.11 Diagnostics for the MARS model.

to GLMs. Hastie et al. (2001) discuss additive models as part of a larger review and compare them to competitive methods.

Exercises

1. The `fat` data gives percentage of body fat, age, weight and height, and 10 body circumference measurements, such as the abdomen, are recorded for 252 men. Body fat is estimated through an underwater weighing technique, but this is inconvenient to use widely. In this question, we develop an additive model that allows the estimation of body fat for men using only a scale and a measuring tape. Your model should predict % body fat according to Siri. You may not use Brozek's % body fat, density or fat-free weight as predictors.
 - (a) Plot the Siri measure against each of the potential predictors and comment on the general pattern of relationships seen in these plots.
 - (b) Fit a linear model with the body fat as the response and use all the remaining variables as predictors. Use Cook Statistics to identify any extremely influential points. Eliminate these points and refit. (You may need to iterate this process.) List the influential points found and identify what particular values contribute to their status. What predictor is most significant in your final model? What value of R^2 is achieved?
 - (c) Fit an additive model for the body fat with smooths on all the available predictors. Identify any influential points.
 - (d) Plot the transformations on the predictors identified by the additive model. Which predictor has the strongest relation to the response? What happens at the extreme values of x observed for some predictors?

- (e) What is the value of R^2 for this additive model and how does it compare to the linear model? Does this mean the additive model is better? Which predictor makes the most nonlinear contribution to the prediction of the response? For this predictor, test whether it can be replaced with a linear term.
 - (f) For the predictor identified in the previous question, plot the transformation used. Describe the nature of the relationship and attempt an interpretation. How can we tell from just the plot that this function is significantly nonlinear?
2. Find a good model for volume in terms of girth and height using the trees data. We might expect that $\text{Volume} = c * \text{Height} * \text{Girth}^2$ suggesting a logarithmic transformation on all variables to achieve a linear model.
- (a) Plot the relationship between the response and each of the predictors.
 - (b) Fit a linear model with all three variables log-transformed. Check the diagnostics of this model. Do the coefficients of the model accord with the theoretically expected relationship?
 - (c) Fit an additive model to the log-transformed variables. Plot the transformations chosen by the default additive model fit. How do these compare to the linear model?
 - (d) Fit an additive model with a log-transformed response but with smooths on the untransformed predictors. Does this additive model point towards the expected log transformations?
 - (e) Fit an ACE model. Plot the transformations found. Do these point towards the logged relationship that theory predicts?
3. The pima dataset consists of 768 female Pima Indians. We want to predict the diabetes test result from the other predictors.
- (a) Plot the test results against each of the predictors. Identify impossible values of the predictors and replace them with the missing value code. Plot the data again and comment on the relationships.
 - (b) Find the subset of cases that are complete (i.e., contain no missing values). From this subset, take a random sample of size 100. This is the test set. All the remaining cases, including those with missing values, are the training set.
 - (c) Fit a GLM to the diabetes test outcome using all the predictors on the training set. Predict the response, negative if $\hat{p} < 0.5$, positive otherwise for the cases in the test set. Make a table showing how the test set predictions compare with the actual outcomes.
 - (d) Use stepwise AIC variable selection on the GLM. Use this reduced model to predict the outcomes, evaluating its performance as in the last question.
 - (e) Fit a GAM using all the predictors. Evaluate the performance on the test set.
 - (f) Fit a GAM using only those predictors selected by the reduced GLM model. Evaluate performance.
 - (g) Compare the results from the four different models. Is one of them clearly better than the others? Would this classification method be suitable for use in practice?

4. The `dvisits` data comes from the Australian Health Survey of 1977–1978 and consist of 5190 single adults where young and old have been oversampled.
 - (a) Build a generalized additive model with `doctorco` as the response and `sex`, `age`, `agesq`, `income`, `levyplus`, `freepoor`, `freerepa`, `illness`, `actdays`, `hscore`, `chcond1` and `chcond2` as possible predictor variables. You will need to decide which variables can reasonably be smoothed. Which variables are not statistically significant in your model.
 - (b) Fit a GAM that does not include the insignificant variables identified in the previous question. Test whether this model can be preferred to the larger model.
 - (c) Fit a GLM with the same predictor set as the previous question. Is this model superior to the GAM?
 - (d) What sort of person would be predicted to visit the doctor the most under the smaller GLM model.
 - (e) If you did the exercise on this data in Chapter 5, compare the interpretations of that selected GLM with the GAM used here.
 - (f) For the last person in the dataset, compute the predicted probability distribution for their visits to the doctor, i.e., give the probability they visit 0, 1, 2 etc. times.
5. The `ethanol` dataset in the `lattice` package presents data from ethanol fuel burned in a single-cylinder engine. The emissions of nitrogen oxides should be considered as the response and engine compression and equivalence ratio as the predictors.
 - (a) Reproduce the example plots given on the help page for `ethanol` to reveal the relationship between the variables.
 - (b) Fit a GAM with a bivariate `s()` on the predictors. Plot the relationship and comment.
 - (c) Refit the GAM but with an appropriate kind of smoothing. Plot the surface and compare.
 - (d) Fit a GAM with univariate smooths on the predictors. Can this model be used in preference to the bivariate version used in the previous question?
 - (e) Fit a MARS model with univariate smooths on the predictors. Compare it to the equivalent GAM fit.
 - (f) Now fit a bivariate MARS model and compare to the GAM version.
 - (g) Plot the relationships found by the MARS model and compare to those seen previously.